

I am a postdoctoral assistant professor at the University of Michigan, working in algebra and algebraic geometry.

Specifically, I am interested in: derived categories of projective varieties and quivers; DG and A_∞ algebras; Bridgeland stability; DM stacks, twisted sheaves and Brauer groups; stable sheaves and moduli spaces; Brill–Noether theory; classical low-dimensional algebraic geometry.

Employment & Education

University of Michigan, Postdoctoral Assistant Professor

Sep. 2024–present

Mentor: Alexander Perry

University of Illinois Chicago, Ph.D. Mathematics

2019–2024

Advisor: Izzet Coskun

Nanjing University, B.S. Mathematics

2015–2019

Research

1. *Phantom categories on 3-vertex directed DG quivers*
arXiv:2609.27191; with Yu Shen
2. *Cartan determinants in algebraic geometry*
arXiv:2609.29598; with Yu Shen and Ziyu Zhang
3. *A counterexample to a global-dimension bound for weighted projective lines*
arXiv:2608.23981; with Bochao Kong and Yu Shen
4. *A counterexample to Han’s conjecture*
arXiv:2608.00177, submitted; with Bochao Kong and Yu Shen
5. *A counterexample to DG version of Han’s conjecture*
arXiv:2512.12460, submitted; with Yu Shen
6. *Picard group action on the category of twisted sheaves*
J. Noncommut. Geom. (2026), published online first; with Ting Gong and Yu Shen
7. *A looming of phantoms*
Adv. Math. 499 (2026), 111046; with Kimoi Kemboi, Daniel Krashen, Tianle Liu, Eoin Mackall, Svetlana Makarova, Alexander Perry, Antonios-Alexandros Robotis, and Sridhar Venkatesh
8. *Geometric phantom categories do not admit Noetherian t -structures*
arXiv:2503.02052, submitted
9. *Stringy Chow rings and weighted blow-ups*
Int. Math. Res. Not. 2026, no. 3, rnaf374; with Qiangru Kuang, Rachel Webb, and Weihong Xu
10. *Morita theory on root gerbes*
arXiv:2409.20317, submitted; with Yu Shen
11. *Stability conditions on surface root stacks*
arXiv:2407.19104, submitted; with Yu Shen
12. *Cones of effective cycles on blow ups of projective spaces along rational curves*
arXiv:2308.13964, submitted; with Benjamin Gould
13. *Spherical vector bundles on nodal $K3$ surfaces*
arXiv:2307.00091, submitted
14. *Nonexistence of exceptional bundles on $\mathbb{P}^3$ with maximal possible ranks*
J. Algebra 695 (2026), 283–310
15. *The cohomology of spherical vector bundles on $K3$ surfaces*
arXiv:2210.11030; preprint being polished
16. *Higher rank Brill–Noether theory on $\mathbb{P}^2$*
Int. Math. Res. Not. 2023, no. 24, 22096–22137; with Benjamin Gould and Woohyung Lee

Awards

UIC Dean’s Scholar Fellowship

Fall 2023

Mentoring

<i>Directed Reading Program on quiver representations</i> Mentees: Yotam Chopra and Xiangnong Wu	Spring 2026
<i>Directed Reading Program on algebraic geometry</i> Mentee: Neelima Borade	Spring 2020

Teaching

University of Michigan Calculus II (Fall 2024), Linear Algebra (Spring 2025), Calculus I (Fall 2025), Multivariable Calculus (Spring 2026), Calculus II (Fall 2026)	2024–2026
University of Illinois Chicago Mathematical Reasoning (Fall 2019), Elementary Linear Algebra (Spring 2020), Calculus I (Fall 2020), Calculus I (Spring 2021), Calculus I (Fall 2021), Calculus II (Spring 2022), Precalculus (Summer 2022), Mathematical Reasoning (Fall 2022), Calculus I (Spring 2023), Calculus II (Summer 2023), Calculus II (Fall 2023)	2019–2023

Selected Talks

<i>Lectures on Han’s conjecture</i> 4-lecture series, Seminario de Álgebra Homológica y Hierbas Afines (SAHHAS), Universidad de Buenos Aires	Fall 2026 (upcoming)
<i>Geometric phantom categories do not admit Noetherian t-structures</i> Nanjing University algebraic K-theory conference	Summer 2026
<i>A counterexample to DG version of Han’s conjecture</i> AMS 2026 Spring Eastern Sectional Meeting, Boston, Massachusetts	Spring 2026
<i>Stability conditions on surface root stacks</i> Caltech/USC joint algebra & geometry seminar, Pasadena, California	Fall 2024
<i>Geometric phantom categories do not admit Noetherian t-structures</i> Brown University algebraic geometry seminar, Providence, Rhode Island	Spring 2025
<i>Nonexistence of exceptional bundles on $\mathbb{P}^3$ with maximal possible ranks</i> Michigan algebraic geometry seminar, Ann Arbor, Michigan	Fall 2023
<i>The cohomology of spherical vector bundles on K3 surfaces</i> BUGCAT 2022, Binghamton, New York	Winter 2022
<i>Higher rank Brill–Noether theory on $\mathbb{P}^2$</i> GTA 2022, Philadelphia, Pennsylvania	Summer 2022

Organized Events

<i>Midwest Algebraic Geometry Graduate Student Conference 2022</i> Co-organized with Sixuan Lou, Shravan Patankar, Benjamin Tighe, and Junyan Zhao	Summer 2022
<i>Derived Category Student Seminar</i> Co-organized with Benjamin Gould	Spring 2022
<i>UIC Graduate Algebraic Geometry Seminar</i> Co-organized with Kuang Yu Wu	Spring 2022
<i>Gromov–Witten Theory Student Seminar</i> <i>UIC Graduate Algebraic Geometry Seminar</i> Co-organized with Kuang Yu Wu	Winter 2021 Fall 2021